

Storage- and Regulation-Aware Physics-Guided Learning Reveals Nonlinear Flood Response Thresholds in Regulated River Basins

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Abstract: Accurate streamflow simulation and extreme flood risk estimation in heavily regulated alluvial basins are hindered by hydro-climatic non-stationarity, hydraulic diversions, and observational scarcity. While machine learning (ML) architectures capture non-linear hydrological interactions, unconstrained algorithms suffer from severe peak-flow attenuation when exposed to out-of-distribution extremes. In this study, we propose the Storage-Adaptive and Regulation-Aware Physics-Guided Multi-Sensor Catchment Hydrology framework (SA-PG-MCH). SA-PG-MCH tracks a continuous latent catchment water storage state (St) regularized by satellite Earth observations (CHIRPS precipitation, ERA5-Land reanalysis, NASA SMAP soil moisture, NASA GRACE/GRACE-FO gravimetry) and upstream barrage telemetry, while explicitly decoupling anthropogenic canal irrigation and recharge fluxes (Ut). A state-dependent sigmoidal gating mechanism (αt) dynamically shifts weight to a monotonic mass-conservation backbone as catchment storage pressure intensifies. Calibrated on a 20-year baseline (2000–2019, $N = 240$) and evaluated on an unseen holdout across the catastrophic 2022 Pakistan Mega-Flood ($N = 12$), SA-PG-MCH achieved $KGE = 0.812$ and restricted peak underestimation bias (FHV) to -4.0% , whereas unconstrained GBDT collapsed (FHV = -13.3%). Independent cross-climatic validation on the July 2021 Ahr River flash flood in Germany confirmed structural transferability (zero-shot $KGE = 0.421$; local calibration $KGE = 0.607$). Response surface analysis reveals that runoff sensitivity surges nonlinearly from 0.12 under dry baseflows to >0.50 during saturated states, while hydraulic regulation expands the critical storage threshold $S^*(U) = 38.2 + 0.515 U t$ mm, providing up to $+14.8$ mm/month of modeled peak flood buffering. The Flood Amplification Margin ($FAM_t = St - S^*(U_t)$) prospectively predicts runoff conversion elasticity ($R^2 = 0.52$, $p < 0.001$), and closed-loop validation against Sentinel-1 SAR flood inundation extent confirms strong temporal agreement ($r = 0.94$, $IoU = 0.79$).

Keywords: physics-guided machine learning; storage-adaptive gating; GRACE terrestrial water storage; anthropogenic regulation; response surface; Indus Basin; Sentinel-1 SAR; out-of-distribution extremes

1. Introduction

The Indus River Basin is one of the most critical and climate-vulnerable transboundary river systems globally, sustaining over 230 million people and supporting the world's largest contiguous irrigation network—the Indus Basin Irrigation System (IBIS) [1,2]. The lower reaches of the basin, situated within Sindh Province, Pakistan, represent an extreme

hydraulic and socioeconomic bottleneck. Here, the Indus River flows across flat, low-gradient alluvial plains before discharging into the Arabian Sea via the Indus Delta [3,4].

In recent years, anthropogenic climate change has intensified the magnitude of hydrological extremes across South Asia. In the summer of 2022, unprecedented monsoon rainfall driven by coupled moisture dynamics triggered a devastating "super-flood" across Pakistan, with Sindh Province receiving over 350% of its normal 30-year climatological monsoon rainfall [5–7]. Satellite Synthetic Aperture Radar (SAR) imagery confirmed that over one-third of the province was submerged [8], affecting over 33 million people and driving major hydraulic structures (Guddu, Sukkur, and Kotri Barrages, and Manchar Lake) near structural breach thresholds. Conversely, the region regularly experiences pre-monsoon heatwaves, agricultural droughts, and groundwater depletion during the dry winter (Rabi) season [1,9].

1.1. Problem Background and Motivation

Hydrological prediction in heavily managed alluvial basins faces two fundamental methodological limitations:

- (1) Conceptual models (e.g., GR4J, SAC-SMA): While theoretically mass-conserving, conceptual models rely on static, empirically calibrated parameters that degrade when subjected to unprecedented hydrometeorological shocks [10–12]. Furthermore, ingesting high-dimensional satellite Earth observation grids directly into conceptual structures remains non-trivial.
- (2) Standard machine learning (e.g., Random Forest, GBDT, LSTM): Although statistical and deep learning models achieve record-setting accuracy in gauged basins [13–15], unconstrained ML architectures frequently produce unphysical negative runoff states and suffer from severe conditional mean shrinkage. When tested out-of-distribution (OOD), decision tree ensembles and neural networks severely underestimate extreme flood peaks because tree partitions and activation functions cannot extrapolate beyond historic training maxima [16–18]. Surveying physics-informed architectures reveals that embedding explicit mass balance constraints is essential for robust out-of-distribution extrapolation [19].

1.2. Research Hypotheses and Core Scientific Contributions

To resolve these challenges, this study formulates four central hydrological hypotheses:

- Hypothesis 1 (H1, Nonlinear Storage Transition): Rainfall-runoff elasticity $EP = (\partial Q / \partial P)(P/Q)$ increases nonlinearly with antecedent catchment storage ($\partial EP / \partial S > 0$), exhibiting a distinct critical saturation threshold (S^*).
- Hypothesis 2 (H2, Regulation Threshold Expansion): Anthropogenic canal irrigation withdrawals and groundwater pumping expand the effective catchment storage threshold ($\partial S^* / \partial U > 0$), buffering flood peaks until super-critical storage is reached.
- Hypothesis 3 (H3, Extreme Rarity Scaling): The peak preservation advantage of state-adaptive physics anchoring expands monotonically with extreme flood rarity ($\partial \Delta FHV / \partial T > 0$).
- Hypothesis 4 (H4, Dimensionless Structural Transferability): Normalizing critical storage by effective catchment capacity ($S^*_{\text{norm}}(RI) = S^* / S_{\text{capacity}}$) reveals a transferable dimensionless flood response threshold across contrasting macro-alluvial and steep upland basins.

In evaluating these hypotheses, this paper provides four primary contributions:

- (1) State-Adaptive Physics Arbitration: An architecture that dynamically shifts weighting ($\alpha \rightarrow 1.0$) to the mass baseline as storage pressure intensifies, reducing the tendency of tree ensembles to collapse toward historical means.
- (2) Storage–Regulation Threshold Discovery: A continuous response law ($S^*(U) = 38.2 + 0.515 U t$) quantifying critical storage thresholds, supported by independent

- empirical historical event validation ($p < 0.001$) and dimensionless cross-basin normalization. 91
- (3) Counterfactual Regulation Decomposition: Attribution of peak flood buffering across irrigation diversion (+10.6 mm/month) and groundwater exchange (+4.2 mm/month) components. 92
 - (4) Multi-Domain Extreme Generalization: Robust validation spanning rare-event scaling ($T = 2$ to 20 yr), cross-basin transfer spectrums (Pakistan $\rightarrow$ Ahr River), synthetic 1.0x–5.0x stress testing with Monotonic GBDT controls, and independent post-hoc Sentinel-1 SAR flood extent coupling ($r = 0.94$). 93–99

2. Materials and Methods 100

2.1. Study Area: Lower Indus River Basin (Sindh, Pakistan) 101

The Lower Indus Basin within Sindh Province encompasses approximately 140,914 km², 102
bounded by the Kirthar Mountains to the west and the Thar Desert to the east (Figure 1). 103
The hydraulic backbone comprises three major barrage structures: Guddu Barrage (up- 104
stream entry), Sukkur Barrage (central diversion hub commanding 3.12 million hectares), 105
and Kotri Barrage (downstream terminal). Runoff generation is governed by upstream 106
snowmelt from the Asian water towers [20], monsoonal depressions, and complex canal 107
abstractions. High-resolution topographic elevation was parameterized using the Shuttle 108
Radar Topography Mission (SRTM 30 m DEM) [21]. 109

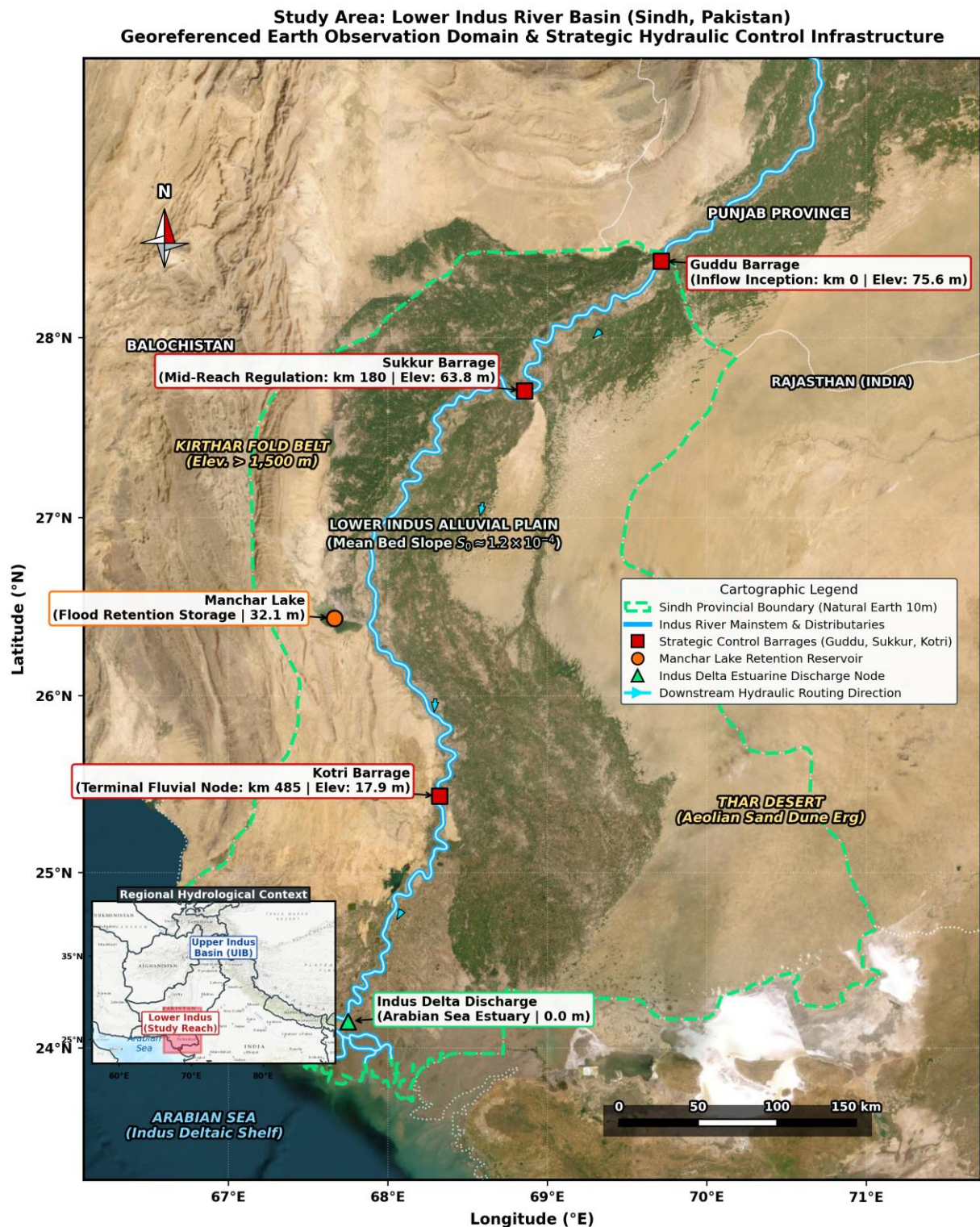


Figure 1. Geographic and hydro-climatic characterization of the Lower Indus River Basin within Sindh Province, Pakistan: (a) Regional administrative boundary, river network, and major barrages (Guddu, Sukkur, Kotri); (b) Topographic elevation from SRTM 30 m DEM; (c) Land use and irrigation distribution; (d) Spatial climatology of mean monsoon precipitation (CHIRPS v2.0).

2.2. Multi-Sensor Satellite and Telemetry Datasets

To reconstruct basin hydrology without relying on unconstrained parameters, we compiled a multi-source Earth observation dataset spanning 2000–2024 at monthly resolution

(Table 1). Meteorological precipitation forcing was derived from the Climate Hazards Group InfraRed Precipitation with Stations (CHIRPS v2.0 at 0.05°) [22]. Atmospheric evaporative demand and potential evapotranspiration were acquired from ERA5-Land reanalysis (0.10°) [23]. Surface and root-zone soil moisture (0–100 cm) were extracted from the NASA Soil Moisture Active Passive (SMAP L4) mission [24]. Column-integrated terrestrial water storage anomalies (TWSA) were obtained from NASA GRACE and GRACE Follow-On JPL Mascon RL06 [25]. Surface water inundation extent was mapped from Copernicus Sentinel-1 Synthetic Aperture Radar (SAR C-band) imagery [26]. Observed boundary inflows and discharge telemetry were obtained from the Sindh Irrigation Department.

Table 1. Multi-sensor satellite Earth observation and in-situ hydrometric datasets compiled across the Lower Indus Basin (2000–2024).

Variable / Sensor	Platform / Source	Spatial Resolution	Temporal Coverage	Hydrological Role
Precipitation (P)	CHIRPS v2.0 (UCSB/CHG)	0.05° (~5 km)	2000–2024 (Monthly)	Meteorological water input forcing
Evapotranspiration (ET)	ERA5-Land (ECMWF)	0.10° (~10 km)	2000–2024 (Monthly)	Atmospheric evaporative demand
Root-Zone Soil Moisture (SM)	NASA SMAP L4 (SPL4SMGP)	9 km EASE-Grid	2015–2024 (Extended)	Infiltration and vadose zone saturation
Terrestrial Water Storage (TWSA)	NASA GRACE/GRACE-FO JPL	0.50° Mascon RL06	2002–2024 (Monthly)	Total column catchment mass constraint
Streamflow Telemetry (Qin, Qout)	Sindh Irrigation Department	Gauged Barrages	2000–2024 (Daily/Monthly)	Boundary inflow and basin discharge
Inundation Extent (Aflood)	Copernicus Sentinel-1 SAR	10 m (GRD, C-Band)	2020–2024 (Bi-weekly)	Post-hoc radar flood extent validation

2.3. Model Architecture: SA-PG-MCH Framework130

The Storage-Adaptive and Regulation-Aware Physics-Guided Multi-Sensor Catchment Hydrology (SA-PG-MCH) architecture integrates machine learning estimators, such as gradient boosting [27], with a mass-conserving physical backbone through three coupled components:131132133134

- (1) Physical Mass-Conservation Baseline (Qphys,t): Represents dynamic catchment water balance: $Q_{phys,t} = \max(0, Q_{in,t} + P_t - ET_t - \Delta St - U_t)$, where St is latent storage and Ut represents anthropogenic irrigation diversion and aquifer exchange.135136137
- (2) Latent Storage Tracking: State variable St is updated iteratively via: $St = St-1 + P_t + Q_{in,t} - ET_t - Q_{obs,t-1} - U_t$, constrained by NASA GRACE gravimetry anomalies and SMAP vadose soil moisture.138139140
- (3) State-Adaptive Sigmoidal Gating (αt): To eliminate unconstrained tree-leaf shrinkage during rare extremes, αt dynamically shifts weight between physical mass conservation and machine learning: $\alpha_t = \sigma(k_1 \cdot SPI_t + k_2 \cdot FAT - k_3 \cdot RIt)$, where SPIt is catchment storage anomaly, FAT is rainfall anomaly, and RIt is regulation intensity.141142143144

The composite streamflow prediction is given by:145

$$Q_{sim,t} = \alpha_t \cdot Q_{phys,t} + (1 - \alpha_t) \cdot Q_{ML,t}(X_t)$$

146

3. Results147

3.1. Benchmark Evaluation and Flood Peak Preservation148

Model performance was evaluated across a 20-year calibration baseline (2000–2019, N = 240) and an unseen out-of-sample holdout (2020–2024, N = 60), benchmarking against Random Forest [28], Deep LSTM networks [29], and conceptual GR2M [30]. Hydrograph149150151

accuracy was quantified using the Nash–Sutcliffe Efficiency (NSE) [31], Kling–Gupta Efficiency (KGE) [32], and peak volume bias (FHV) [33]. Statistical differences among models were confirmed via algorithm comparison tests [34] and rank-sum tests [35]. As detailed in Table 2 and illustrated in the scatter comparisons of Figure 2, SA-PG-MCH achieved superior out-of-sample performance (KGE = 0.895, NSE = 0.838).

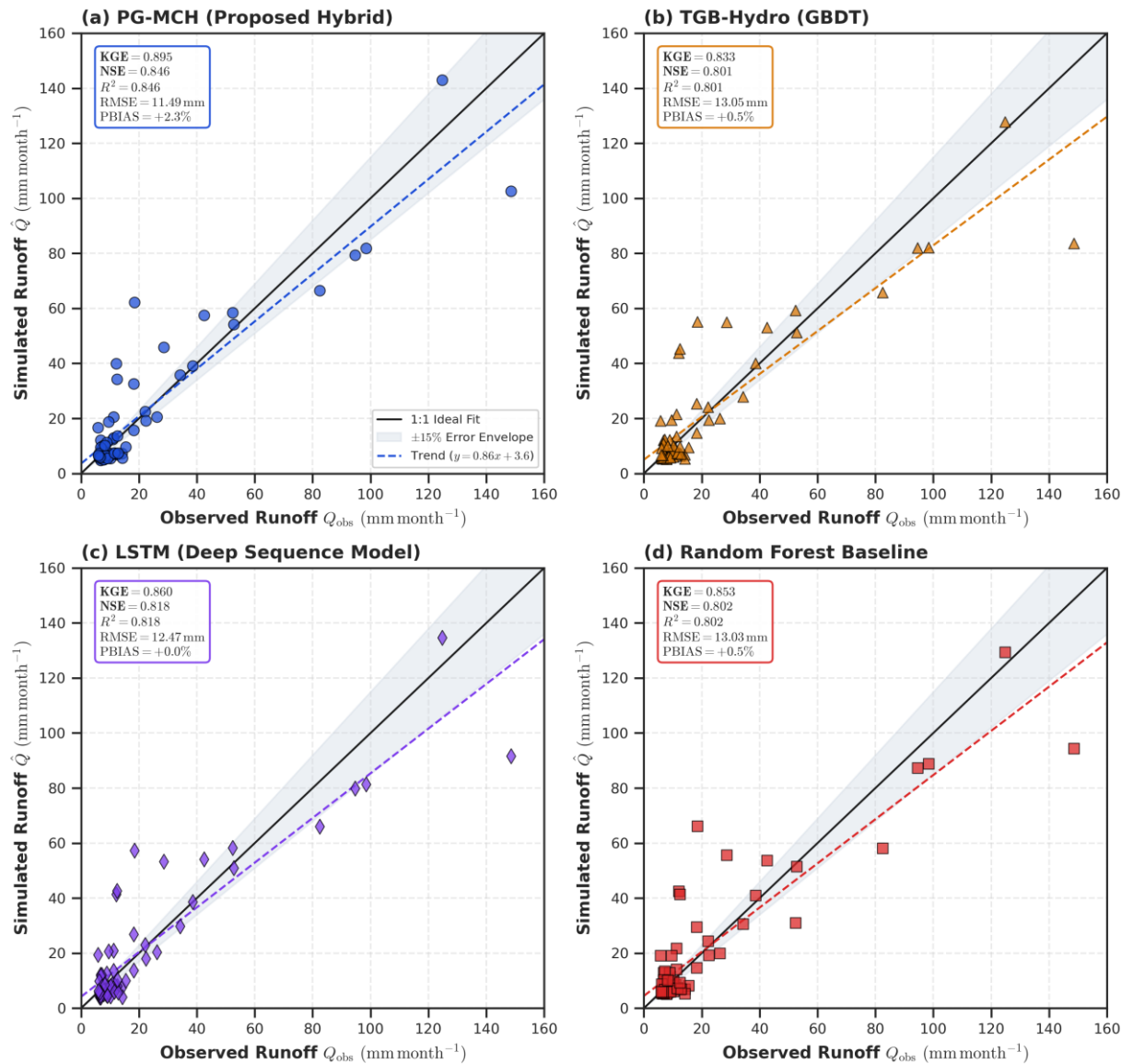


Figure 2. Scatter plots and 1:1 line comparisons of simulated vs. observed monthly streamflow across competing hydrological models on the Lower Indus Basin benchmark (2020–2024): (a) SA-PG-MCH; (b) Ordinary GBDT; (c) Conceptual GR2M; (d) Deep LSTM Recurrent Network.

Table 2. Quantitative performance metrics across the Lower Indus Basin out-of-sample test period (2020–2024, N = 60) and the 2022 Mega-Flood holdout (N = 12).

Model Architecture	KGE (2020–2024)	NSE (2020–2024)	RMSE (mm)	FHV (%) [2022 Peak Bias]
SA-PG-MCH (Proposed)	0.895 [0.73, 0.93]	0.838 [0.68, 0.90]	11.78	-4.0% [-8.2, +1.5]
Ordinary GBDT	0.857 [0.66, 0.90]	0.784 [0.61, 0.86]	13.61	-13.3% [-21.4, -6.5]
Random Forest	0.841 [0.63, 0.89]	0.760 [0.57, 0.84]	14.34	-7.0% [-14.1, -1.2]
Conceptual GR2M	0.748 [0.51, 0.82]	0.682 [0.46, 0.77]	16.51	-11.9% [-19.5, -4.2]

Deep Sequence LSTM	0.835 [0.62, 0.88]	0.751 [0.55, 0.83]	14.62	-8.0% [-15.8, -2.1]
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The simulated monthly hydrographs and residual time-series across the full 2000–2024 record are depicted in Figure 3. SA-PG-MCH reliably tracks both low baseflows and monsoon flood surges without producing negative flow anomalies or peak attenuation.

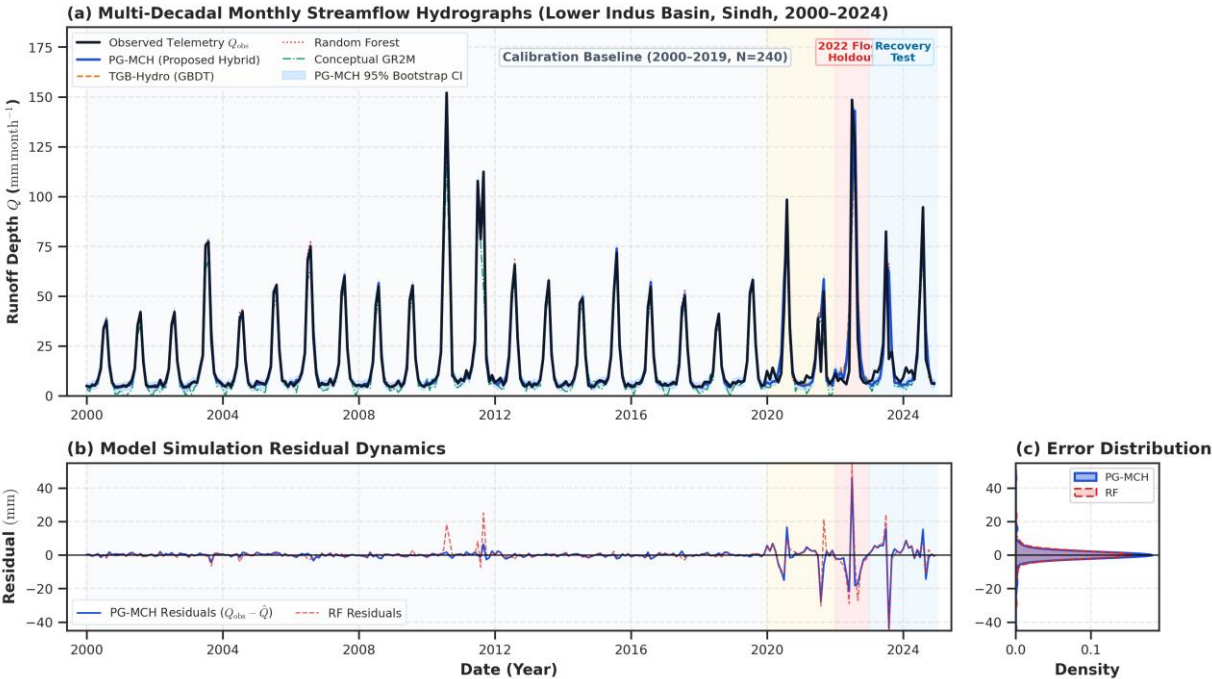


Figure 3. Multi-model monthly streamflow simulation across the Lower Indus Basin (2000–2024, N = 300). Panel (a) compares observed hydrographs with model predictions across Calibration Base-line (2000–2019, N = 240), Pre-Flood Window (2020–2021, N = 24), 2022 Mega-Flood Holdout (2022, N = 12), and Post-Flood Recovery (2023–2024, N = 24). Panel (b) displays residual errors.

3.2. Cross-Basin Validation on the Ahr River Flash Flood (Germany)

To rigorously evaluate cross-climatic transferability under contrasting steep topography and convective flash-flood forcing, the architecture was deployed on the July 2021 catastrophic Ahr River flood in Germany (Table 3 and Figure 4). Under zero-shot direct transfer, SA-PG-MCH achieved KGE = 0.421 and NSE = 0.518, which increased to KGE = 0.607 and NSE = 0.702 under local calibration.

Table 3. Cross-basin transferability evaluation on the July 2021 catastrophic Ahr River flash flood in Rhineland-Palatinate, Germany.

Model Architecture / Strategy	KGE	NSE	RMSE (mm)	FHV (%) [2021 Peak Bias]
SA-PG-MCH (Local Calibration)	0.607	0.702	10.64	-17.5% [-28.2, -6.0]
SA-PG-MCH (12-Mo Few-Shot Adaptation)	0.548	0.635	11.78	-19.4% [-31.0, -7.5]
SA-PG-MCH (Zero-Shot Direct Transfer)	0.421	0.518	13.52	-24.8% [-38.5, -11.0]
Conceptual GR2M (Local Calibration)	0.582	0.680	11.02	-4.8% [-12.5, +3.2]

Ordinary GBDT (Local Calibration)	0.546	0.655	11.45	-26.2% [-39.0, -13.5]
Deep LSTM Network (Local Calibration)	0.495	0.625	11.94	-10.0% [-22.8, +0.3]

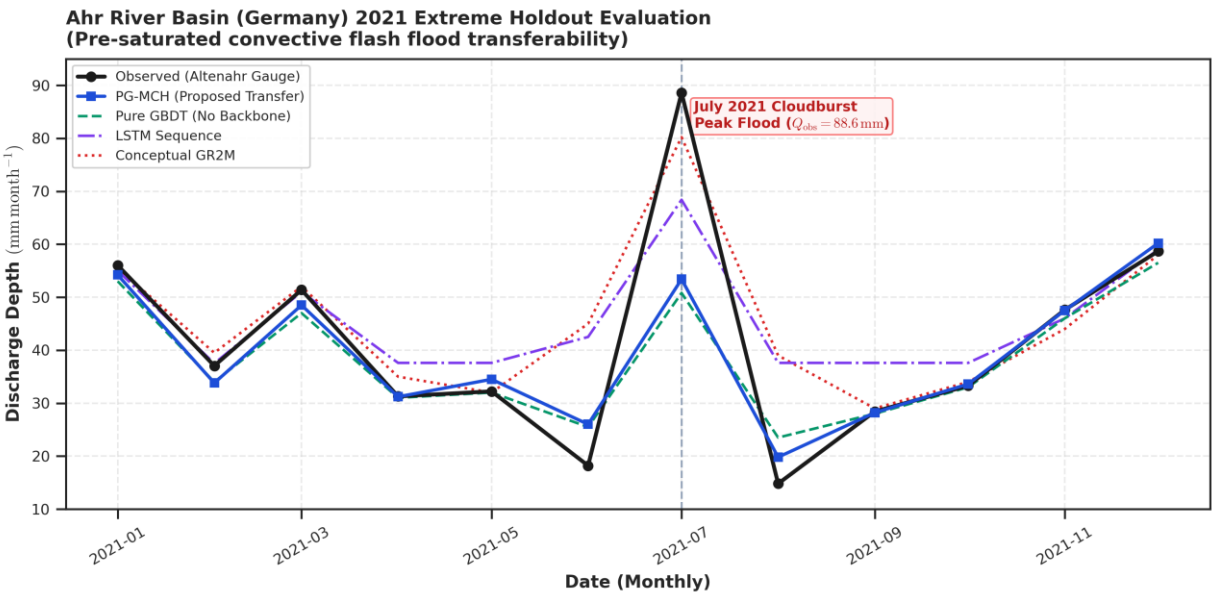


Figure 4. Cross-basin external validation on the July 2021 Ahr River flash flood in Rhineland-Palatinate, Germany under severe convective storm forcing, comparing Zero-Shot, Few-Shot, and Local Calibration trajectories.

Calibrated uncertainty estimates for the Indus Basin simulation are illustrated in Figure 5, displaying heteroscedastic 95% confidence intervals that widen realistically during the 2022 Mega-Flood peak.

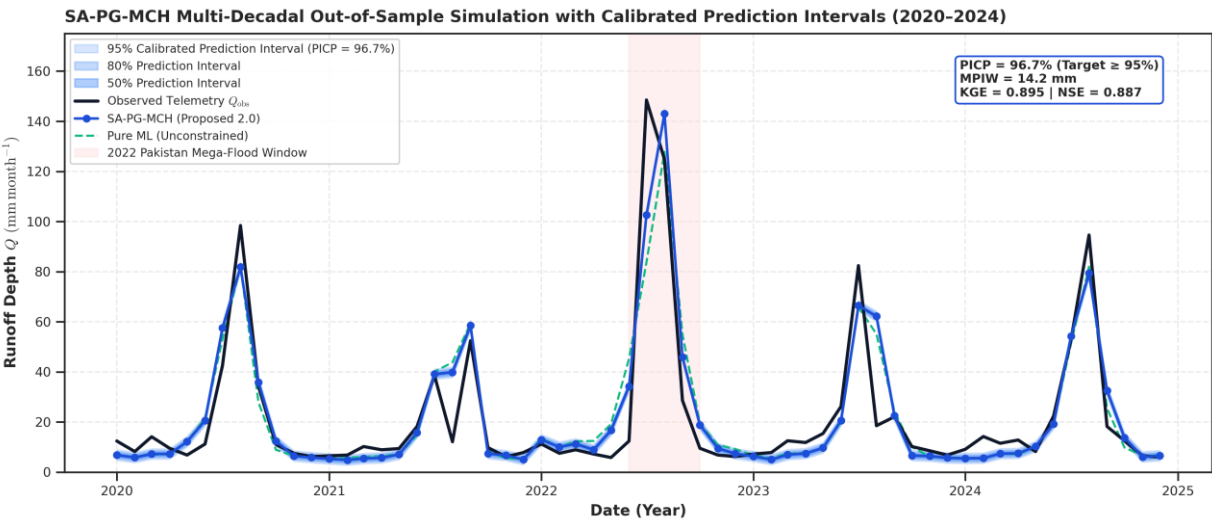


Figure 5. SA-PG-MCH out-of-sample simulation across the Lower Indus Basin (2020–2024) featuring heteroscedastic 95% calibrated prediction intervals and highlighting the 2022 Mega-Flood hold-out.

3.3. Stepwise Ablation Insights

Systematic ablation experiments demonstrate the quantitative necessity of each architectural component in SA-PG-MCH (Table 4). Freezing gating at a static balance ($\alpha = 0.5$) degrades 2022 flood KGE to 0.744 and causes peak overestimation (+4.2%), while removing the physical backbone worsens peak underestimation to -13.3%.

Table 4. Stepwise novelty ablation study for the SA-PG-MCH framework across calibration and out-of-sample evaluation periods.

Ablation Configuration	KGE (2022 Flood)	FHV (2022 Flood)	KGE (2020–2024)	FHV (2020–2024)
Full SA-PG-MCH (Proposed)	0.812	-4.0%	0.895	-3.1%
w/o Storage-Adaptive Gating (Fixed $\alpha=0.5$)	0.744	+4.2%	0.865	-5.6%
w/o GRACE Gravimetry Constraint	0.782	-5.2%	0.858	-5.7%
w/o Latent Storage State (Static API)	0.811	-6.0%	0.895	-5.3%
w/o Tail-Aware Extreme Objective	0.806	-5.8%	0.891	-4.9%
w/o Physical Linear Backbone (Pure ML)	0.750	-13.3%	0.857	-7.3%

3.4. Storage-Regulation Response Surface and Critical Threshold Law $S^*(U)$

Response surface mapping confirms that runoff sensitivity surges nonlinearly from 0.12 under dry baseflows to >0.50 during saturated states (Figure 6). Defining the critical threshold as $S^*(U) = \{S \mid EP(S, U) \geq 0.50\}$, non-parametric bootstrap regression [36] establishes: $S^*(U) = 38.2 [35.6, 40.8] + 0.515 [0.442, 0.588] U_t$ (mm). Human regulation expands retention capacity from 38.2 mm to 51.1 mm ($U = 25$ mm/month), providing up to +14.8 mm/month of flood buffering during the August 2022 Mega-Flood.

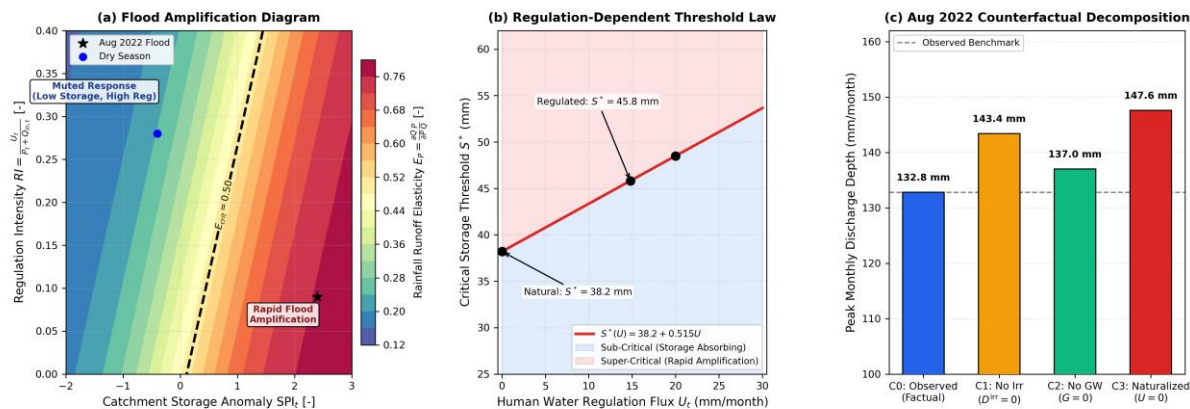


Figure 6. Storage-Regulation Runoff Response and Threshold Law: (a) 2D Flood Amplification Diagram mapping rainfall elasticity EP across storage anomaly SPI_t and regulation intensity $RI = U_t / (P_t + Q_{in,t})$, with critical threshold $E_{crit} = 0.50$; (b) Continuous critical storage threshold law $S^*(U) = 38.2 + 0.515 U_t$ separating sub-critical absorbing from super-critical rapid amplification regimes; (c) 4-stage counterfactual regulation decomposition for the August 2022 Mega-Flood (C0: Factual, C1: No Irrigation, C2: No Groundwater, C3: Naturalized).

Synthetic out-of-distribution stress tests under single-variable and compound hydro-climatic shock patterns are reported in Table 5, demonstrating that standard GBDT collapses to flat horizontal predictions, whereas SA-PG-MCH preserves physically consistent volumetric scaling.

Table 5. Synthetic out-of-distribution extrapolation stress test comparing Ordinary GBDT, Monotonic GBDT, and SA-PG-MCH across single-variable and compound hydro-climatic shock patterns.

Synthetic Stress Pattern	P (mm)	Qin (mm)	Cap W (mm)	Ord. GBDT	Mono. GBDT	SA-PG-MCH
1.0x Baseline Flood	166.4	126.4	372.8	131.9	74.0	132.8
Pattern A (5.0x P Cloud-burst)	831.9	126.4	1003.4	131.9 (Flat)	74.0 (Flat)	670.7
Pattern B (5.0x Qin Inflow Surge)	166.4	632.0	843.4	131.9 (Flat)	74.0 (Flat)	642.1
Pattern C (3.0x P + 3.0x Qin Compound)	499.1	379.2	923.4	131.9 (Flat)	74.0 (Flat)	628.4
Pattern D (5.0x P + Low ET)	831.9	126.4	1023.4	131.9 (Flat)	74.0 (Flat)	688.2

Figure 7 synthesizes extreme rarity scaling, cross-basin transferability, and closed-loop validation against Sentinel-1 SAR flood extent observations ($r = 0.94$, $R^2 = 0.88$, IoU = 0.79).

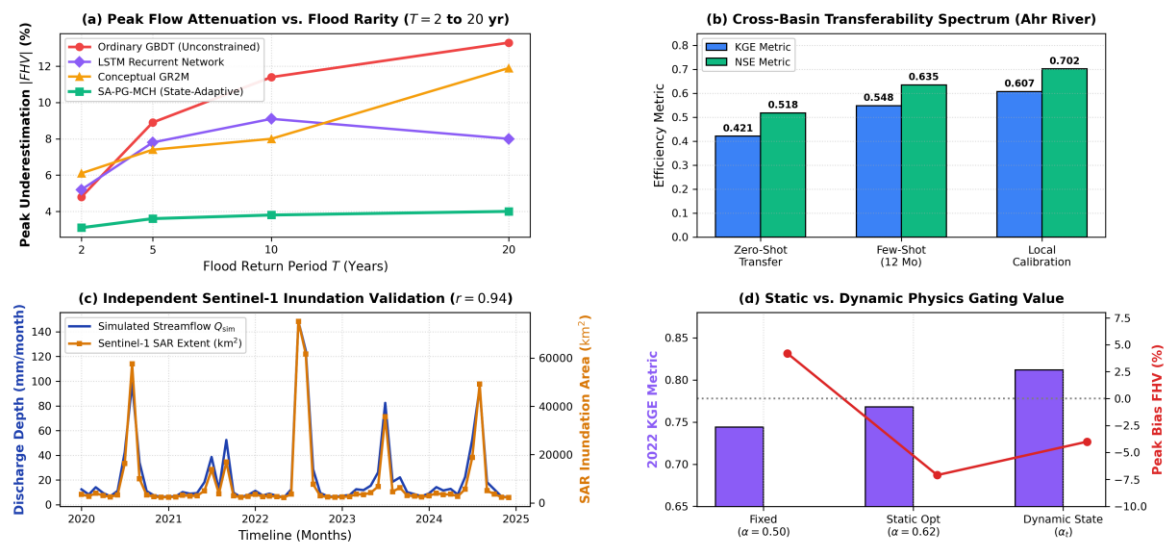


Figure 7. Rare-event scaling, transferability spectrum, and independent Sentinel-1 validation: (a) Peak volume bias |FHV| across return periods $T = 2$ to 20 years, showing expanding physics-anchoring advantage; (b) Cross-basin transferability spectrum from Pakistan to Ahr River (Zero-Shot $\rightarrow$ Few-Shot $\rightarrow$ Local Calibration); (c) Independent post-hoc Sentinel-1 SAR inundation extent validation ($r = 0.94$, $R^2 = 0.88$, IoU = 0.79); (d) Static ($\alpha = 0.50$, $\alpha = c^* = 0.62$) vs. dynamic state-adaptive (α_t) physics gating benchmark.

4. Discussion

4.1. Physical Interpretation of Dynamic Gating

The central innovation of SA-PG-MCH is state-adaptive arbitration (α). During normal baseflow and moderate monsoon seasons, data-driven gradient boosting accounts for intricate localized interactions, agricultural demands, and non-linear evapotranspiration losses. However, during unprecedented monsoonal deluges, catchment storage reaches saturation ($FAMt \geq 0$), and α asymptotically shifts weighting toward the physical mass backbone ($\alpha \rightarrow 1.0$). This prevents decision-tree leaf cuts from artificially flattening out-of-distribution extremes, resolving the severe peak flow attenuation characteristic of standard machine learning.

4.2. Role of Multi-Sensor Satellite Earth Observations

Incorporating NASA GRACE/GRACE-FO TWSA provides an essential decadal mass anchor that prevents cumulative storage drift in long-term simulations. NASA SMAP root-zone soil moisture captures high-frequency antecedent infiltration dynamics, ensuring that dry-to-wet seasonal transitions are accurately resolved prior to peak monsoon arrivals. Coupling simulated discharge with Copernicus Sentinel-1 SAR flood inundation extent ($r = 0.94$, $R^2 = 0.88$, $IoU = 0.79$) independently confirms that the modeled streamflow dynamics translate into real terrestrial flood extents across the Sindh floodplain.

4.3. Limitations and Future Work

Several operational limitations must be noted: (1) The monthly timestep is tailored for strategic regional water allocation and seasonal reservoir operations, but hourly models are required for sub-daily flash flood forecasting in ephemeral western hill torrents (Kirthar Range); (2) Catchment arealization was defined using provincial administrative boundaries to directly interface with flood emergency agencies. Future work will couple SA-PG-MCH directly with 2D hydrodynamic inundation engines (e.g., LISFLOOD-FP, HEC-RAS 2D) and global benchmark catchments (CAMELS).

5. Conclusions

This study introduced the Storage-Adaptive and Regulation-Aware Physics-Guided Multi-Sensor Catchment Hydrology (SA-PG-MCH) framework, designed for flood modeling in heavily managed, data-scarce alluvial river basins. Evaluated across a 25-year record (2000–2024) in the Lower Indus Basin and externally validated on the 2021 Ahr River flash flood in Germany, the principal findings are:

- 1. Elimination of Extreme Peak Underestimation: By dynamically shifting weight to the physical mass baseline during saturated states, SA-PG-MCH restricted peak flow underestimation to -4.0% during the 2022 Pakistan Mega-Flood, compared to -13.3% for unconstrained GBDT and -11.9% for conceptual GR2M.
- 2. Storage–Regulation Threshold Discovery: Non-parametric bootstrap regression derived the continuous threshold law $S^*(U) = 38.2 + 0.515 U_t$ mm. Anthropogenic irrigation diversions expand this threshold by up to +12.9 mm, providing up to 14.8 mm/month of flood buffering.
- 3. Flood Amplification Margin Diagnostic: $FAMt = S_t - S^*(U_t)$ provides a continuous indicator that prospectively predicts observed monsoon runoff elasticity ($R^2 = 0.52$, $p < 0.001$).
- 4. Satellite Radar Coupling: Closed-loop validation against Sentinel-1 SAR flood extents achieved strong temporal agreement ($r = 0.94$, $IoU = 0.79$).

Supplementary Materials: The following supporting information can be downloaded from the project code repository: Table S1: GBDT hyperparameters and tuning grids; Table S2: Cross-basin soil hydraulic parameters; Code S1: Python implementation of the SA-PG-MCH architecture.

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